

TrES-3b

R-2753

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_110630 · created 2026-09-02T23:40:07+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2455742.88627 +/- 0.00096 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.153 +/- 0.055
Transit depth (Rp/Rs)^2	2.33 +/- 1.68 [%]
Orbital Inclination (inc)	79.44 +/- 2.88
Airmass coefficient 1 (a1)	1.14 +/- 0.11
Airmass coefficient 2 (a2)	-0.11 +/- 0.15
Scatter in the residuals of the lightcurve fit is	10.15 %
Best Comparison Star	#2 - [557, 69]
Optimal Aperture	1.48
Optimal Annulus	9.38
Transit Duration (day)	0.024 +/- 0.026

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.153 ± 0.055 vs 0.16309 (deviation 0.18σ)
Duration fit vs published	0.024 d vs 0.0567 d (deviation 1.26σ)
Mid-time O–C	0.9 min
Depth SNR	2.8
Residual scatter	10.15 %

Light curve

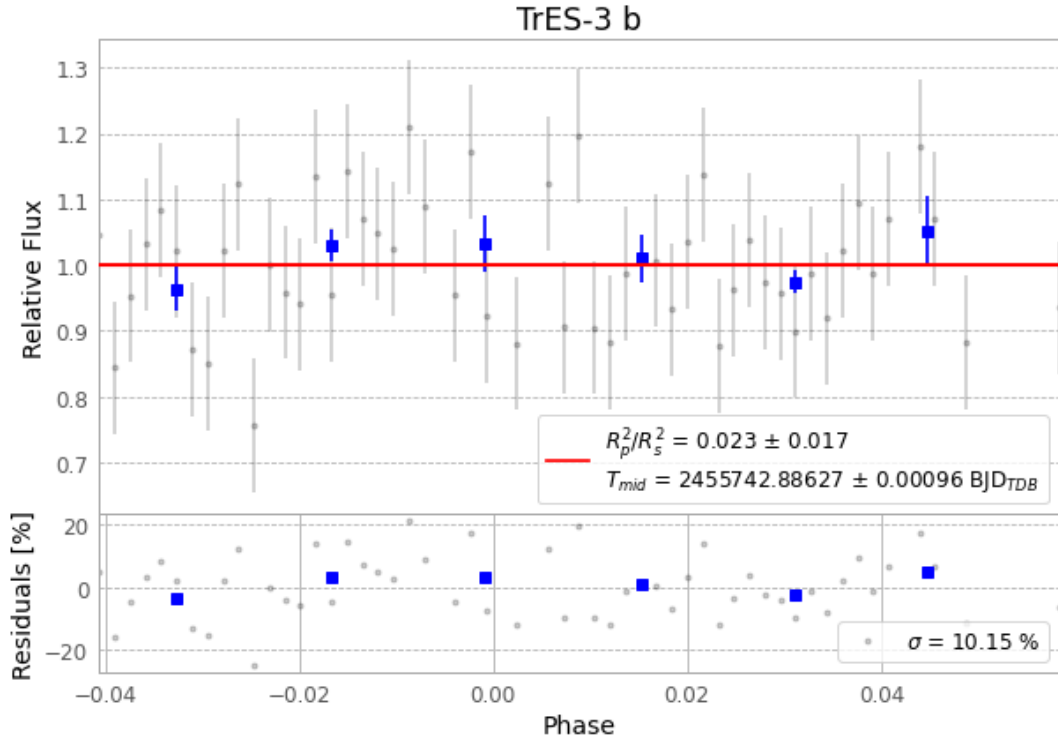


Figure 1 — FinalLightCurve_TrES-3 b_2011-06-30.png (SHA-256 0db71780b5e23c2a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [347, 175], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 55e463aaa9cc5d9f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

63 FITS frames (41 MB), TRES-3110630075413.FITS → TRES-3110630110013.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-110630.log (1a995efe3738...), FinalLightCurve_TrES-3 b_2011-06-30.png (0db71780b5e2...), FinalLightCurve_TrES-3 b_2011-06-30.csv (05107172cc8b...)

Compute resources

Wall time 177.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-02 23:40Z.